

Exact optimality of a packing of seven disks of radii $1, 2, \dots, 7$ in a circle

Proof note — independent verification draft

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Abstract

We prove that the minimum radius of a circle containing pairwise disjoint disks of radii $1, \dots, 7$ is the unique root $R_0 \in (13.46211, 13.46212)$ of a five-angle closure equation. The global lower bound involves only the largest five disks: all but one cyclic order are excluded by five elementary rational angle-sum certificates. The remaining order is excluded below R_0 by a strict radial-cosine corner maximum. Two smaller disks fit at fixed rational coordinates. The auxiliary numerical inequalities are reproduced by a standalone exact-rational Python checker; no floating-point result is used as a proof decision.

1 Main theorem and critical value

Write $p_i \in \mathbb{R}^2$ for the center of the disk of radius i . A packing in a container of radius R means

$$\|p_i\| + i \leq R, \quad \|p_i - p_j\| \geq i + j \quad (i \neq j). \quad (1)$$

For $R > i + j$, put

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{ij}{(R-i)(R-j)}} \in (0, \pi). \quad (2)$$

Let $\mathcal{C} = (7, 3, 5, 6, 4)$, with cyclic successor pairs $E_{\mathcal{C}} = \{(7, 3), (3, 5), (5, 6), (6, 4), (4, 7)\}$, and define

$$\Phi(R) = \sum_{(i,j) \in E_{\mathcal{C}}} \beta_{ij}(R), \quad \Phi(R_0) = 2\pi. \quad (3)$$

Every β_{ij} is strictly decreasing in R on $(13, 14)$. Exact rational arctangent series show $\Phi(13.46211) > 2\pi$ and $\Phi(13.46212) < 2\pi$, so (3) defines a unique R_0 in that interval. Numerically $R_0 \simeq 13.4621106776084$.

Theorem 1. *The minimum enclosing-circle radius for seven disks of radii $1, 2, \dots, 7$ is exactly R_0 defined by (3).*

2 A lower bound using only five large disks

Suppose, for contradiction, that a packing exists with $R < R_0$. It is also a packing in the larger rational container of radius

$$U = 13.46212 > R_0.$$

Set $s_i = \|p_i\|$. By containment and non-overlap,

$$0 < s_i \leq u_i = U - i, \quad s_i \geq \ell_i = \max\left(0, \max_{j \neq i}(i + 2j - U)\right), \quad 3 \leq i \leq 7. \quad (4)$$

The lower bound follows from $i + j \leq s_i + s_j$ and $s_j \leq U - j$. All five intervals have strictly positive lower ends.

Lemma 1 (Four-corner maximum). *For $d > 0$ and x, y in a strictly positive rectangle, the maximum of*

$$c_d(x, y) = \frac{x^2 + y^2 - d^2}{2xy}$$

is attained at one of the four corners.

Proof. For fixed y , the derivative $\partial_x c_d = (x^2 - y^2 + d^2)/(2yx^2)$ can change sign only from negative to positive; thus a maximum in x is attained at an endpoint. The same argument on both resulting edges applies in y . $\square$

For each pair of the five largest disks, the cosine law gives a necessary smaller central-angle bound

$$\gamma_{ij} \geq \varphi_{ij}(s_i, s_j), \quad \varphi_{ij}(x, y) = \arccos c_{i+j}(x, y). \quad (5)$$

In particular, every directed angular gap between two consecutive disk centers is at least φ_{ij} . Applying Lemma 1 to the rational radial intervals (4), the following rational ticks q_{ij} are certified lower bounds for φ_{ij} :

Pair	q_{ij}	Pair	q_{ij}	Pair	q_{ij}
(3, 4)	0.5479	(3, 5)	0.8484	(3, 6)	1.0016
(3, 7)	1.1822	(4, 5)	1.0466	(4, 6)	1.2447
(4, 7)	1.4864	(5, 6)	1.5208	(5, 7)	1.8545
(6, 7)	2.4066				

Each tick is independently checked by comparing the exact rational four-corner maximum of c_{i+j} with a rational Taylor lower bound for $\cos q_{ij}$. Therefore no numerical inverse trigonometric functions are needed for the lower-bound certificate.

Up to reflection and rotation, the four directions of disks 4, 5, 6, 7 have three cyclic orders. Two immediately contradict the total 2π angular budget, using the sums of their four consecutive ticks:

Cyclic order	Sum of rational lower ticks
(7, 4, 5, 6)	$6.4604 > 44/7 > 2\pi$
(7, 5, 4, 6)	$6.5524 > 44/7 > 2\pi$

The only possible order (up to reflection) is (7, 4, 6, 5). Disk 3 has a nonzero center radius and a distinct center direction: indeed all its angle ticks against the four others are positive. There are exactly four ways to insert its direction into this order. Three more are impossible:

Cyclic order	Sum of rational lower ticks
(7, 3, 4, 6, 5)	$6.3501 > 44/7 > 2\pi$
(7, 4, 3, 6, 5)	$6.4112 > 44/7 > 2\pi$
(7, 4, 6, 3, 5)	$6.4356 > 44/7 > 2\pi$

The sole surviving order is (7, 4, 6, 5, 3), the reversal of $\mathcal{C}$.

It remains to exclude this order when $R < R_0$. For each of its five adjacent pairs $(i, j) \in E_{\mathcal{C}}$, the exact verifier establishes the following *strict* wall-corner property, uniformly for every $V \in [13.46211, 13.46212]$:

$$c_{i+j}(V - i, V - j) > c_{i+j}(x, y) \quad \text{at each of the other three corners of } [\ell_i(V), V - i] \times [\ell_j(V), V - j], \quad (6)$$

where $\ell_i(V) = \max(0, \max_{k \neq i}(i + 2k - V))$. By the derivative argument of Lemma 1, the wall corner is then the unique global maximum on the rectangle. For the hypothetical packing with $R < R_0$, $s_i \leq R - i < R_0 - i$; hence for each of the five adjacent pairs,

$$\varphi_{ij}(s_i, s_j) > \varphi_{ij}(R_0 - i, R_0 - j) = \beta_{ij}(R_0). \quad (7)$$

Summing the five directed gap inequalities around the surviving cyclic order and using (3) yields

$$2\pi \geq \sum_{(i,j) \in E_{\mathcal{C}}} \varphi_{ij}(s_i, s_j) > \sum_{(i,j) \in E_{\mathcal{C}}} \beta_{ij}(R_0) = 2\pi,$$

a contradiction. Thus every seven-disk packing has $R \geq R_0$.

3 An exact matching upper witness

Set $a_i = R_0 - i$ for $i = 3, 4, 5, 6, 7$. Starting at $\theta_7 = 0$, place the large disk centers at

$$p_i = a_i(\cos \theta_i, \sin \theta_i), \quad \theta_{c_{k+1}} = \theta_{c_k} + \beta_{c_k, c_{k+1}}(R_0).$$

The closure equation (3) makes all five successive large disk pairs tangent, including the last-to-first pair. The other five large-disk pairs have strictly positive certified squared-distance slack. Place the two remaining disks at the *rational* centers

$$p_1 = \left(\frac{193}{20}, -\frac{151}{20} \right), \quad p_2 = \left(\frac{687}{100}, -\frac{907}{100} \right). \quad (8)$$

The exact verifier checks the two containment inequalities, all five nonadjacent large-disk pairs, and all eleven pairs incident to at least one of disks 1, 2. All are strictly satisfied. The minimum certified squared noncontact separation slack exceeds 1; the two small disks' containment squared slacks are also greater than 1. Thus a packing exists at $R = R_0$, establishing the reverse inequality $R^* \leq R_0$. This proves Theorem 1.

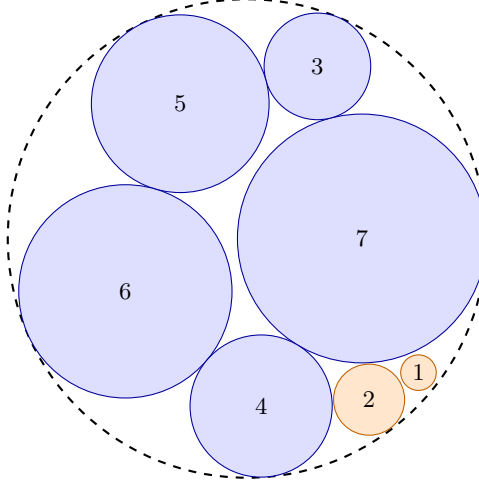


Figure 1: Illustration of the exact seven-disk witness. The blue disks form the rigid five-disk ring. Only the equations and the rational certificates, not these rounded coordinates, are used in the proof.

4 Reproducibility and proof boundary

The accompanying standalone file `certify_integer_seven.py` uses `fractions.Fraction` and integer square roots to compute outward rational intervals. It verifies the rational root bracket by Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and alternating arctangent series with explicit remainder bounds; it checks all angle ticks by a rational cosine Taylor lower bound. It then replays all five angular exclusion cases, all five strict wall-corner comparisons, and all noncontact/containment inequalities of the witness. The geometric lemmas above are mathematical arguments, while finite inequality checks rely on the correctness of the short Python verifier. This is a computer-assisted proof note, not a formalization in Lean. No novelty or priority claim is made without a separate literature review.